

Khaled GASSARA

Software Engineer

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1 year experience · Full-stack and automotive software development · Stack: Python, React, Node.js, Django, PostgreSQL

PROFESSIONAL SUMMARY

Software Engineer with hands-on experience in Python and JavaScript, building reliable full-stack systems for automotive, web, and real-time monitoring platforms. Skilled in test automation and modern frameworks such as React, Node.js, and Django, with a focus on scalable, cloud-ready solutions.

SKILLS

LANGUAGES

Python · JavaScript · TypeScript · Java · SQL · PHP · C#

BACKEND

Node.js · Django · Express.js · REST APIs · Django REST Framework · Flask

DEVOPS & CLOUD

Docker · Kubernetes · AWS · Jenkins

SECURITY

Trivy

MOBILE

Flutter

TOOLS

Git

FRONTEND

React · CSS · Angular · Tailwind CSS

DATABASES

PostgreSQL · MongoDB

QUALITY & TESTS

SonarQube

DATA & AI

Machine Learning · RAG

METHODOLOGIES

GitOps

OTHER

WebSockets · CAN Protocol · LIN Protocol · SOME/IP · MQTT · OpenCV · Deep Learning · LLM Agents · scikit-learn · Unity

LANGUAGES

Arabic · **Native** · English · **Fluent** · French · **Fluent**

EDUCATION

Ingénieur – Software Engineering

National School of Computer Science (ENSI)

06/2025

Engineering Studies

Preparatory Institute for Engineering Studies of Sfax (IPEIS)

07/2022

PROJECTS

AfricaRice

02/2025 – 03/2025

- Developed a cross-platform rice quality analysis app with Flutter, integrating a custom computer vision model (**85%+** accuracy)
- Designed and trained the ML model to analyze rice quality from images
- Built a custom tokenizer/compression approach, shrinking the model from **2 GB** to **~400 MB** for low-end devices
- Optimized the app and inference pipeline for efficient on-device AI on resource-constrained hardware

Dart · Flutter · Python · Computer Vision · Machine Learning · Deep Learning · Image Processing · Model Optimization · Tokenization

Secure CI/CD Pipeline

10/2024 – 11/2024

- Designed and deployed a secure CI/CD pipeline automating delivery of containerized apps with Docker, Jenkins, and Kubernetes
- Integrated SonarQube static analysis and Trivy image scanning for quality and security gates
- Used GitOps for configuration management, reducing deployment time and eliminating manual steps

Docker · Jenkins · SonarQube · Trivy · Kubernetes · GitOps

github.com/KhaledGs5

Fixprosthodontics

08/2024 – 10/2024

- Designed and developed a cross-platform app (Windows & Android) with Flutter for managing dentistry student assessments, targeting teaching staff
- Delivered from prototype to production, covering UX design, backend implementation, and deployment
- Used CSV for lightweight data handling, enabling offline access and simple import/export

Flutter · CSV

Ninga Game

06/2024 – 07/2024

- Designed and developed a 2D game in Unity with C#, featuring AI-powered gesture recognition for real-time character control

- Used Python for gesture recognition models, enabling intuitive input without controllers

Python · Unity · C#

Voice-controlled Assistant with Image Pipeline

05/2023 – 07/2023

- Built a voice-enabled assistant using Python, SpeechRecognition, and pyttsx3 for speech-to-text and text-to-speech
- Integrated retrieval-augmented lookup and ML models (scikit-learn) for context-aware responses
- Owned the full stack: backend orchestration, NLP model integration, and client-facing voice interface

Python · SpeechRecognition · pyttsx3 · scikit-learn

WORK EXPERIENCE

Python Developer

10/2025 – PRESENT

KPIT · Sfax, Tunisia

- Developed **50+** Python-based automated test solutions for automotive ECUs, validating communication across CAN, LIN, and SOME/IP protocols.
- Optimized Python test execution to run **10+** automated tests daily, shortening validation cycle time and accelerating ECU testing.
- Developed solutions using the MTF framework, supporting **100+** automated test scenarios for automotive ECU validation.
- Configured an AI-powered development agent with custom skills to automate **10+** development and testing tasks, reducing manual effort and accelerating engineering workflows.

Full Stack Intern

02/2025 – 06/2025

Telnet

- Developed and integrated full-stack features using React and Node.js, delivering scalable functionality to **600+** users.
- Developed resource tracking and session management features using Express.js and MongoDB, streamlining training operations and data management.
- Developed RESTful APIs using Node.js and Express.js, handling **10,000+** data records daily and enabling seamless frontend-backend integration.
- Coordinated workflows for **600+** employees, streamlining task management and improving operational efficiency.

Full Stack Intern (Part-Time)

07/2024 – 08/2024

Primatec Engineering

- Built a real-time vehicle monitoring app with React and Django, streaming live data to **100+** connected systems
- Implemented WebSocket-based live updates for low-latency vehicle tracking
- Integrated Mosquitto MQTT broker to handle real-time feeds from **100+** connected systems
- Streamlined testing workflows, accelerating operations for **100+** workers and boosting testing efficiency

Test Automation Intern

06/2024 – 07/2024

Telnet

- Developed **30+** automated test suites with Robot Framework and Python, reducing manual testing effort
- Implemented multithreaded test execution, reducing test execution time by **40%** and improving overall testing throughput
- Created **30+** reusable test scripts, reducing test duplication and accelerating regression cycles

Full Stack Intern

07/2023 – 08/2023

Cristal Laboratory at ENSI

- Built a full-stack web app with React frontend and PHP/SQL backend
- Designed database schemas and implemented **20+** RESTful API endpoints for core application features
- Styled responsive UI components with CSS, ensuring cross-device compatibility

INTERESTS

Artificial Intelligence · Gaming · Open Source